

Total Marks: 30

TSE, Spring 26.

Time: 30 Minutes

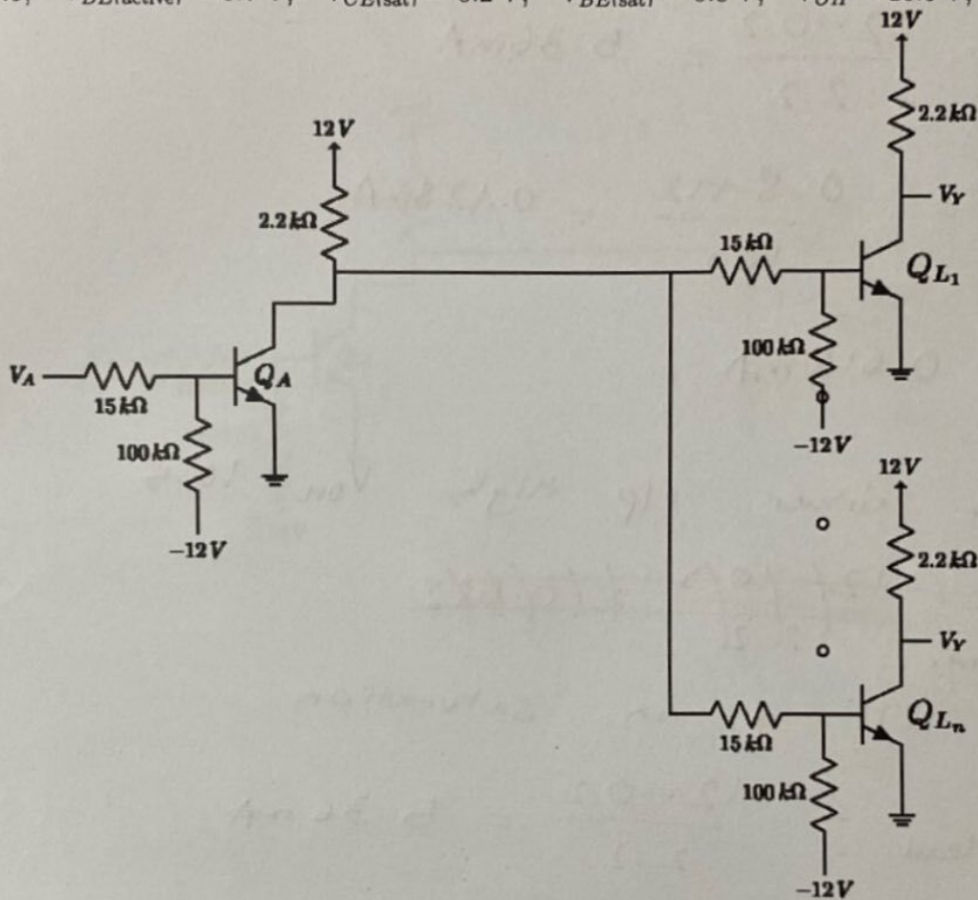
Name:

ID:

Section:

1. For the circuit given,

$$\beta = 27.5, \quad V_{BE(\text{active})} = 0.7 \text{ V}, \quad V_{CE(\text{sat})} = 0.2 \text{ V}, \quad V_{BE(\text{sat})} = 0.8 \text{ V}, \quad V_{OH} = 10.5 \text{ V}, \quad V_{OL} = 0.1 \text{ V}$$



- State the operating mode of the transistor Q_A when V_A is 12 V. [2 mark]
- Find the output voltage V_{O1} and I_{15k} , I_{100k} , I_B , I_C for the driver circuit when V_A is 12 V. [8 marks]
- Find the noise margin. [5 marks]
- Find the maximum fanout. [6 marks]
- For the maximum fanout case, find the maximum power dissipation of the full circuit (Driver and loads). [4 marks]

1a) Saturation.

b) When $V_A = 12V$,
 Q_A is in saturation, so, $V_{O_1} = 0.2V$

Since $V_{BE} = 0.8V$,

$$I_{15k} = \frac{12 - 0.8}{15} = 0.746mA$$

$$I_C = \frac{12 - 0.2}{2.2} = 5.36mA$$

$$I_{100k} = \frac{0.8 + 12}{100} = 0.128mA$$

$$I_B = 0.618mA$$

c) When driver o/p High, $V_{OH} = 10.5$

~~$$I_C = \frac{12 - 10.5}{2.2} = 0.682$$~~

Load BJT is in saturation

$$I_{C_{load}} = \frac{12 - 0.2}{2.2} = 5.36mA$$

$$I_B = \frac{I_C}{\beta} = 0.1949mA$$

$$I_B = I_1 - I_2$$

$$\Rightarrow 0.1949 = \frac{V_{IH} - 0.8}{15} - \frac{0.8 + 12}{100}$$

$$\therefore V_{IH} = 5.644V$$

To find V_{IL}

$$\frac{V_{IL} - 0.5}{15} = \frac{0.5 + 12}{100}$$

$$V_{IL} = 2.375$$

$$V_{NH} = 10.5 - 5.644 = 4.856V$$

$$V_{NL} = 2.375 - 0.1 = 2.275V$$

$$NM = \min(4.856, 2.275)$$

$$\boxed{= 2.275V}$$

d) Max Fanout when driver o/p V_{OH}

$$I_c = 0.682$$

$$I_L = \frac{10.5 - 0.8}{15} = 0.6466$$

$$\text{Fanout} = \frac{0.682}{0.6466} = \lfloor 1.055 \rfloor = 1$$

Fanout for driver o/p Low,

$$I_c = \frac{12 - 0.1}{2.2} = 5.41$$

$$I_L = \frac{0.1 + 12}{115} = 0.105$$

$$\text{Fanout} = \lfloor 51.5 \rfloor = 51$$

$$\text{Max} = \min(51, 1) = 1$$

e) Power for max output ≈ 1

When input = 0V

Driver in cutoff, $V_o = 10.5$

$$I_1 = 0.1044 \text{ mA}$$

$$I_c = \frac{12 - 10.5}{2.2} = 0.6818$$

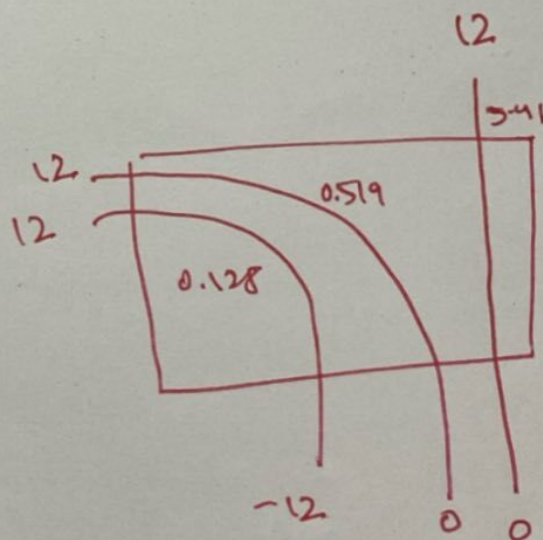
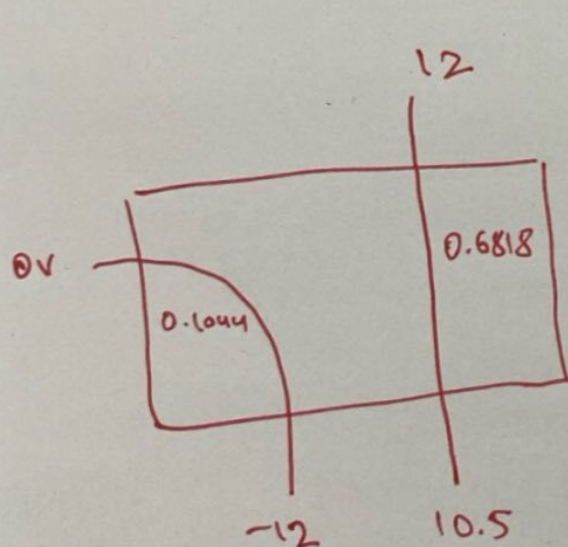
Load in saturation

$$I_1 = \frac{10.5 - 0.8}{15} = 0.647$$

$$I_2 = \frac{12 - 8}{100} = 0.128$$

$$I_B = 0.519 \text{ mA}$$

$$I_c = \frac{12 - 0.1}{2.2} = 5.41$$

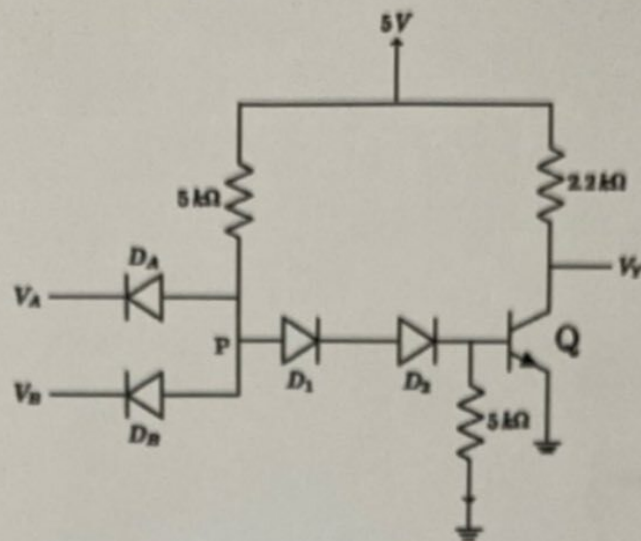


Driver

$$P = (12 \times 0.1044) + (12 - 10.5) \times 0.6818 \\ + (24 \times 0.128) + (12 \times 0.519) + (12 \times 5.41) \\ = 76.49 \text{ mW}$$

2. For the circuit given,

$$\beta = 30, \quad V_{BE(\text{active})} = 0.7 \text{ V}, \quad V_{CE(\text{sat})} = 0.2 \text{ V}, \quad V_{BE(\text{sat})} = 0.8 \text{ V},$$



- (a) For case (1,0) state the operating modes of the diodes D_A , D_B , D_1 , D_2 and the transistor Q . [2 marks]
 (b) Find all voltage and currents for case (1,0). [3 marks]

1a) $D_A \rightarrow \text{RB}$
 $D_B \rightarrow \text{FB}$
 $D_1, D_2, \rightarrow \text{RB}, \quad Q \rightarrow \text{cutoff}$

b) $I_{5k} = \frac{5 - 0.7}{5} = 0.86 \text{ mA}$

All else are 0.